EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 9A

1. I bet Cal will win this year

As huge fans of the Big Game, you and your friend want to bet on whether Cal or Stanford will win this year. You want to predict this year's result by analyzing historical records. Therefore, you decide to model this as a binary classification problem and do PCA for dimension reduction on the data you collected. The "+1" class represents victories of Cal and "−1" represents victories of Stanford.

After some research, you obtained a data matrix $\mathbf{A} \in \mathbb{R}^{d\times n}$,

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \tag{1}$$

where each of the n columns $\mathbf{a}_i$ denotes a game and each of the d rows of $\mathbf{A}$ contains information of a possibly relevant factor of the games (weather, location, date, air quality, etc).

(a) Let the full SVD of $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$. You project your data along $\mathbf{u}_1$ and $\mathbf{u}_2$ (the first two principal components), and for comparison you also project your data along two randomly chosen directions $\mathbf{w}_1$ and $\mathbf{w}_2$ as well. You get the two pictures in Figure 1, but you forgot to label the axes. Of the two figures below, which one is the projection onto the principal components and which one is the projection onto the random directions? **Match axes (i), (ii), (iii), (iv) to $\mathbf{w}_1, \mathbf{w}_2, \mathbf{u}_1$, and $\mathbf{u}_2$, and justify your answer.** *Note that there may be multiple correct matchings; you only need to find and justify one of them.*

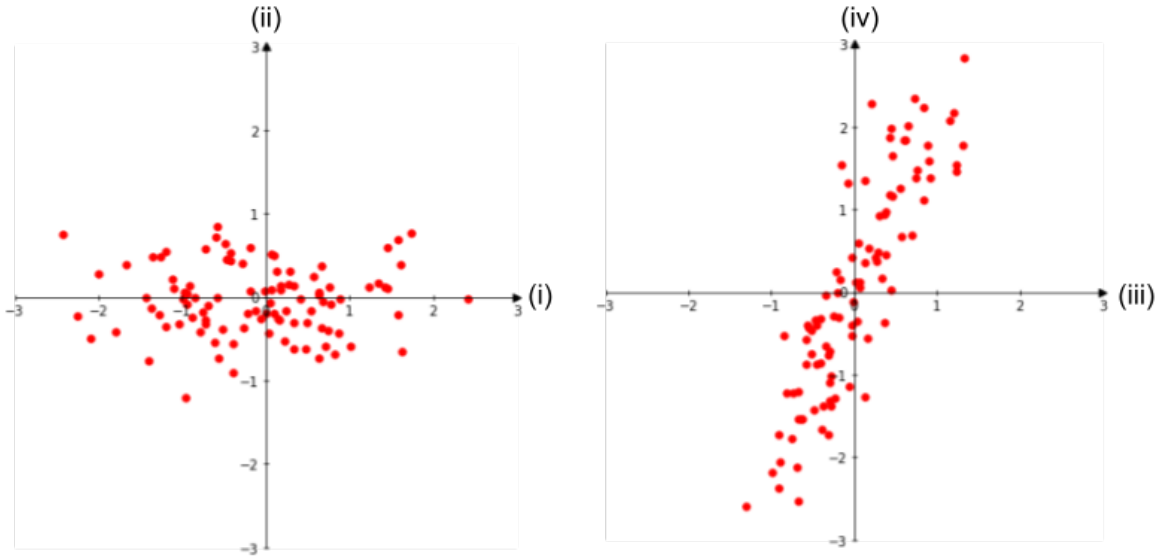


Figure 1: Projected datasets.

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(b) In order to reduce the dimension of the data, we would like to project the data onto the first k principal components of $\mathbf{A}$, where k is less than the original data dimension d. **Show how to find the new vector $\mathbf{z}_i \in \mathbb{R}^k$ which is the k-dimensional, compressed version of $\mathbf{x}_i$.** You may use the SVD of $\mathbf{A}$.

(c) Given a new set of projection coefficients denoted $\mathbf{z}_{\text{new}} \in \mathbb{R}^k$, we can define a classifier that will predict $+1$ (i.e., that Cal wins) if $\mathbf{w}_\star^\top \mathbf{z}_{\text{new}} > 0$ and -1 (i.e., that Stanford wins) otherwise.

Assume $d = 6$, $k = 4$, and $\mathbf{w}_\star = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}^\top$. Let $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$ for $\mathbf{A}$, and you find that $\mathbf{U}$ is given by the identity matrix, i.e. $\mathbf{U} = \mathbf{I}_d$. Now suppose the data point for this year's big game $\mathbf{x}_{2025} = \begin{bmatrix} 3 & 6 & 4 & 1 & 9 & 6 \end{bmatrix}^\top$. **Would you bet on Cal or Stanford to win? Justify your answer.**
(HINT: Don't forget to project your data onto the principal components.)

2. Robotic Grasping with PCA (Adapted from EECS 16A Fa24 Exam 3)

As a robotics automation engineer, we would like to develop a robotic system that can pick up ellipse-shaped objects from a table, using only an overhead camera image as input.

In order to estimate the pose of our elliptical objects, we will characterize it by their center point (x, y), while their orientation is described by a unit vector pointing in the direction of their major axis(the longer axis).

To derive these positions, we will sample points in the image that appear to lie on the object. By doing so uniformly across the image, we assume that they are roughly representative of the object's actual position. An example plot is as follows:

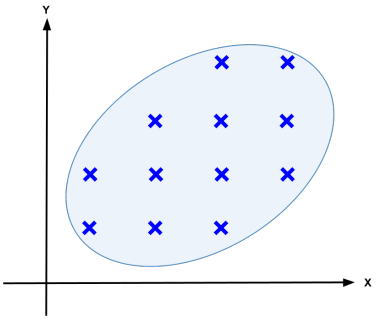


Figure 2: Example of Object Sampling

For one particular sampled object, assume that we store the (x, y) positions of our sampled points as **row data** in the following matrix:

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 1 & 2 \end{bmatrix}$$

(a) Given the data matrix $\mathbf{A}$, estimate the position $(\bar{x}, \bar{y})$ of the center of the ellipse.

(b) Ignoring the previous subpart, imagine that the data has now been centered about the origin, and can now be characterized by the following data matrix $\mathbf{A}'$.

$$\mathbf{A}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ -2 & 2 \\ 2 & -2 \\ -1 & -1 \end{bmatrix}$$

Given this matrix, determine the best estimate for the unit vector $\hat{\mathbf{m}}$ that characterizes the direction of the ellipse's major axis. This axis should correspond to the **direction of maximum variance** in our sampled points.

Now that we have successfully estimated the pose of our target object, we can actually pick it up use a gripper as shown below!

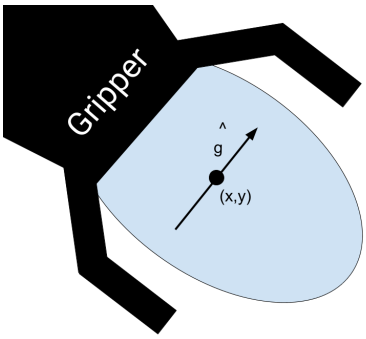


Figure 3: Grasp Visualization

(c) In order to maximize our chances of successfully grasping the object, however, we want to grasp along the **shortest** axis of our ellipse estimation. Assuming that the estimate of the dirrection of ellipse's major axis is $\hat{\mathbf{m}} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$, provide the unit vector $\hat{\mathbf{g}}$ along the **shortest** axis.